

Gianfranco Romani

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PROFESSIONAL SUMMARY

Senior ML Engineer working at the intersection of AI and cybersecurity, focused on both bringing security to AI systems and applying AI to defensive cybersecurity challenges. Designs closed-loop agent environments where models investigate, triage, and remediate vulnerabilities against deterministic verifiers, authors internal agent SDKs, and builds security tooling including Code Scan MCP and the open-source codeflow-solver graph optimization engine. Background combines empirical ML research and multi-GPU fine-tuning (U.S. Patent 12,651,016, EMNLP, WMT) with practical AI security, automated red teaming with Promptfoo, enterprise incident response, and offensive security training (OSCP candidate).

WORK EXPERIENCE

Thomson Reuters

Senior ML Engineer, AI Security & Cybersecurity

Zug (CH), Oct 2025 – Present

- **Remediation Agents & Code Scan MCP:** Architected containerized SAST and SCA remediation agents and built Code Scan MCP to ingest multi-source static analysis findings directly into developer action spaces (presenting at AgentCon Europe 2026). Engineered closed-loop verification environments combining root-cause triage, targeted code patching, and sandboxed build and test execution to validate fixes. Deployed across Thomson Reuters to ~8,000 developers, achieving a >70% PR merge rate for automated security fixes.
- **Internal Agent SDK & Enterprise Standards:** Developed an internal Agent SDK to scale secure agent engineering and enforce production best practices across the organization. Standardized core capabilities across the agent lifecycle, establishing shared frameworks for systematic evaluation and hypothesis testing, end-to-end observability and cost accounting, and runtime security guardrails.
- **Enterprise Agent Safeguards & AI Security:** Defined defense-in-depth controls and threat modeling baselines for agent deployments, mitigating prompt injection, unsafe tool execution, untrusted data flows, and sensitive data leakage. Embedded runtime guardrails and deterministic output validation across production agents.
- **Security Reviews & Incident Response (On-Call AI SME):** Conducted rigorous security reviews across architecture, threat models, and codebases for enterprise AI systems. Served as AI SME during high-impact security incident response, deploying AI-assisted diagnostics to rapidly investigate failure modes and generate verified remediation patches under tight operational timelines.
- **Workshop Delivery (AMLD 2026 @ EPFL):** Delivered a hands-on AI security workshop to ~80 attendees at Applied Machine Learning Days on prompt-injection analysis, model vulnerabilities, and automated red teaming.

Applied ML Scientist II

Zug (CH), Oct 2024 – Oct 2025

- **Adversarial Red Teaming & Tooling Strategy (Promptfoo):** Evaluated open-source and vendor security tooling against standardized criteria, establishing vendor partnership and building custom extensions around Promptfoo. Scaled automated red teaming across all enterprise teams with LLM components, cutting evaluation turnaround from weeks to a few days while uncovering numerous critical vulnerabilities prior to production.
- **Graph Optimization for Vulnerabilities (codeflow-solver):** Developed and open-sourced codeflow-solver, formulating SARIF vulnerability dataflow graphs as a Set Cover Problem to mathematically guarantee the minimum number of code modifications required to eliminate all security findings. [\[GitHub\]](#)
- **Applied GenAI PoCs & RAG:** Delivered multiple high-quality proofs-of-concept spanning autonomous task agents, advanced RAG architectures, and document intelligence pipelines for enterprise workflows.
- **Reading Group Leadership & Technical Writing:** Led the TR Labs Paper Reading Group across Zug, London, and Bangalore offices to discuss and debate frontier ML research; author of the "Essential NLP Reads" series on Medium. [\[Medium\]](#)

Applied ML Scientist I & Applied Research Intern in NLP

Zug (CH), Aug 2023 – Oct 2024

- **Trade Compliance Classification SOTA (U.S. Patent):** Researched and built core ML/NLP models for international trade compliance classification, outperforming commercial SOTA baselines by 20% (presented at EMNLP 2024; foundational work for U.S. Patent No. 12,651,016).
- **Automated GenAI Data Pipelines:** Designed and implemented automated GenAI data processing pipelines, reducing manual curation effort by 90%.
- **Red Teaming Tooling:** Built automated red-teaming and safety testing prototypes for LLM-based applications to evaluate and mitigate emerging risks.

SafEcho AG

Data Science Intern

Remote, May 2023 – Jul 2023

- **Occupational Health Agents:** Researched domain-specific conversational agents for Occupational Safety and Health.
- **Synthetic Data & Tabular QA:** Engineered synthetic data generation pipelines for fine-tuning via OpenAI APIs; built tabular Q&A retrieval systems with Haystack.

Pi School & Translated

AI Fellow

Rome (IT), Mar 2023 – Apr 2023

- **Adaptive MT with LLMs:** Collaborated with Translated on applied research in adaptive machine translation using Large Language Models.
- **Distributed PEFT Fine-Tuning:** Fine-tuned BLOOM and Flan-T5 architectures using Parameter-Efficient Fine-Tuning (PEFT/LoRA) on multi-GPU, multi-node clusters.
- **Constrained Prompt Engineering:** Engineered constrained prompting strategies enforcing strict style, terminology, and domain consistency (published at WMT23).

PATENTS

- Gholamian S., et al. (incl. **Romani G.**), “Systems and methods for harmonized product classification”, **U.S. Patent No. 12,651,016**, granted June 9, 2026 (Assignee: Thomson Reuters). [\[Patent Link\]](#)

PUBLICATIONS

- Sina G., **Romani G.**, Rudnikowicz B., Skylaki L., “LLM-Based Robust Product Classification in Commerce and Compliance”, Workshop on Customizable NLP at **EMNLP 2024**, Miami. [\[Paper Link\]](#)
- Kamal M., **Romani G.**, Ricciuti G., Anagnostopoulos A., Chatzigiannakis I., “Analyzing Topic Models: A Tourism Recommender System Perspective”, **AINA 2024**, Japan. [\[Paper Link\]](#)
- Moslem Y., et al., “Domain Terminology Integration into Machine Translation: Leveraging Large Language Models”, Proceedings of **WMT23 Conference**, Dec 2023. [\[Paper Link\]](#)

CERTIFICATIONS

- **Offensive Security Certified Professional (OSCP), In Progress** (OffSec, PEN-200). Hands-on network penetration testing, Active Directory exploitation, privilege escalation, lateral movement, custom exploit modification, and vulnerability analysis across simulated enterprise cyber ranges and CTF-style target environments.
- **Lakera 101 AI Security:** AI threats, LLM red teaming, prompt injection defenses, OWASP Top 10 for LLMs, AI governance.

EDUCATION

Sapienza Università di Roma

MSc in Computer Science Engineering (Artificial Intelligence & Machine Learning)

Rome (IT), Nov 2020 – Jan 2023

- **Thesis:** A Review of Topic Modeling Techniques and Experimental Evaluation in Analysis of Touristic Experience.

Sapienza Università di Roma

BSc in Computer and Systems Engineering

Rome (IT), Oct 2017 – Oct 2020

- **Thesis:** Application and Optimization of YOLO Tiny for Object Detection in Duckietown Environment.

TECHNICAL SKILLS & TOOLS

- **Machine Learning & AI Research:** Python, PyTorch, Hugging Face (Transformers, PEFT), Scikit-learn, LLM Fine-Tuning (PEFT/LoRA, Multi-GPU), Closed-Loop Agent Environments (Deterministic Verifiers, Sandboxed Execution), RAG, Agent SDKs & Evaluation.
- **Cybersecurity & AI Security:** Security Reviews (Architecture, Threat Modeling, Code Review), Vulnerability Remediation (SAST, SCA), Graph Optimization (Set Cover), Adversarial Red Teaming (Promptfoo), Prompt Injection & Jailbreaking, Network & Active Directory Pentesting (OSCP / PEN-200), Incident Response, OWASP Top 10 for LLMs.
- **Systems, Tools & Infrastructure:** Model Context Protocol (MCP), Langfuse, OpenTelemetry, Linux (Debian/Kali), Docker, Git, CI/CD (GitHub Actions), AWS (SageMaker, S3, AgentCore), PostgreSQL.
- **Languages & Interests:** Fluent in English and Italian, basic in German. European Masters Swimming competitor (2022); Red Cross Volunteer.